

Getting started with Amazon Virtual Private Cloud (Amazon VPC)

AWS Prescriptive Guidance

Amazon Virtual Private Cloud (Amazon VPC) lets you provision a logically isolated section of the AWS Cloud where you can launch AWS resources in a virtual network that you define. This guide covers VPC fundamentals, subnets, route tables, internet gateways, NAT, security groups, network ACLs, VPC peering, Transit Gateway, PrivateLink, and network design best practices for building secure, scalable, and resilient AWS network architectures.

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Introduction

Amazon VPC is the foundational networking layer for almost every AWS service. It provides you with complete control over your virtual networking environment, including resource placement, connectivity, and security. With Amazon VPC, you can define IP address ranges using CIDR blocks, create subnets, configure route tables, and set up both internet-facing and internal-only architectures.

Every AWS account comes with a default VPC in each Region, configured with a /16 IPv4 CIDR block, a default subnet in each Availability Zone, an internet gateway, and a default route table. For production workloads, you should create custom VPCs with carefully planned CIDR ranges, subnet structures, and security configurations.

VPC Fundamentals

When creating a VPC, you assign it an IPv4 CIDR block, which can range in size from /16 (65,536 IP addresses) to /28 (16 IP addresses). You can also associate one or more secondary CIDR blocks and IPv6 CIDR blocks with the VPC. Choose CIDR ranges that do not overlap with your on-premises networks or other VPCs that you plan to connect.

Within a single AWS Region, a VPC spans all Availability Zones. You then create subnets within Availability Zones to place your resources. Each subnet must exist within one AZ and cannot span multiple AZs, providing fault domain isolation.

Note: AWS reserves five IP addresses in each subnet: the network address, the VPC router address, the DNS server address, a reserved address for future use, and the broadcast address.

Subnets

Subnets divide your VPC's CIDR block into smaller segments that you associate with specific Availability Zones. A well-designed subnet architecture is critical for security, fault tolerance, and routing flexibility. The most common pattern is a three-tier architecture with public, private application, and private data subnets.

Subnet Type	Internet Access	Examples	Typical Resources
Public	Yes (via IGW)	10.0.1.0/24 per AZ	Load balancers, NAT Gateways, bastion hosts
Private (App)	Outbound only (via NAT)	10.0.11.0/24 per AZ	EC2, ECS, Lambda in VPC, EKS nodes
Private (Data)	None	10.0.21.0/24 per AZ	RDS, ElastiCache, OpenSearch, EFS

Route Tables and Gateways

Each subnet in a VPC must be associated with a route table. The route table contains rules that determine where network traffic from your subnet is directed. The main route table is created automatically with a VPC and controls routing for all subnets that are not explicitly associated with a custom route table.

Internet Gateway

An Internet Gateway (IGW) is a horizontally scaled, redundant, and highly available VPC component that allows communication between resources in your VPC and the internet. An IGW serves two purposes: to provide a target in your route tables for internet-routable traffic, and to perform network address translation (NAT) for instances with public IPv4 addresses.

NAT Gateway

A NAT Gateway enables instances in a private subnet to initiate outbound traffic to the internet while preventing inbound connections from the internet. NAT Gateways are managed by AWS, are highly available within an AZ, and scale automatically up to 100 Gbps. For high availability, deploy one NAT Gateway per Availability Zone and configure private subnets in each AZ to route internet-bound traffic to the NAT Gateway in the same AZ.

Security Groups and Network ACLs

Amazon VPC provides two levels of network security: security groups and network access control lists (ACLs). Understanding the difference between them is essential for designing a layered defense strategy.

Feature	Security Groups	Network ACLs
Scope	Instance level	Subnet level
State	Stateful (return traffic allowed)	Stateless (return traffic must be explicitly allowed)
Rules	Allow rules only	Allow and Deny rules
Rule evaluation	All rules evaluated	Rules evaluated in order (lowest number first)
Default behavior	Deny all inbound; allow all outbound	Allow all inbound and outbound
Use case	Primary traffic filter	Coarse-grained subnet protection

VPC Connectivity Options

Amazon VPC provides multiple options for connecting your VPC to other networks, including other VPCs, on-premises data centers, and the AWS global network.

VPC Peering

VPC peering enables you to route traffic between two VPCs using private IPv4 or IPv6 addresses. Peered VPCs behave as if they are in the same network. VPC peering can be established between VPCs in the same AWS account, different accounts, and different AWS Regions. VPC peering is non-transitive — if VPC A is peered with VPC B, and VPC B is peered with VPC C, traffic cannot flow from VPC A to VPC C through VPC B.

AWS Site-to-Site VPN

AWS Site-to-Site VPN creates an encrypted tunnel between your on-premises network and your VPC over the public internet. Each VPN connection consists of two tunnels for redundancy. Virtual Private Gateways (VGW) or Transit Gateways serve as the AWS-side endpoint for VPN connections. AWS Accelerated VPN uses AWS Global Accelerator to route VPN traffic over the AWS global network backbone, improving throughput and latency.

AWS Direct Connect

AWS Direct Connect establishes a dedicated network connection from your data center or colocation facility to AWS. This private connection can reduce network costs, increase bandwidth throughput, and provide a more consistent network experience compared to internet-based connections. Direct Connect supports connection speeds from 50 Mbps to 100 Gbps through Direct Connect Partners.

AWS Transit Gateway

AWS Transit Gateway is a network transit hub that connects VPCs, VPN connections, and Direct Connect gateways through a central hub, simplifying your network and reducing the operational complexity of managing many VPC peering connections. Transit Gateway supports transitive routing — all attached VPCs and on-premises networks can communicate through the Transit Gateway without direct peering.

Transit Gateway route tables allow you to segment your network into isolated routing domains, enabling fine-grained control over inter-VPC communication. You can associate multiple VPCs with a single route table (shared services) or use separate route tables for different environments (dev, staging, prod) to enforce network isolation.

AWS PrivateLink

AWS PrivateLink provides private connectivity between VPCs, AWS services, and on-premises applications without exposing traffic to the public internet. PrivateLink uses VPC Interface Endpoints (powered by AWS Hyperplane) to create elastic network interfaces in your subnets with private IP addresses that serve as entry points for traffic destined for supported AWS services or your own service endpoints.

PrivateLink is the recommended pattern for private access to AWS services such as S3, DynamoDB, KMS, SNS, SQS, and many others. It is also used for sharing services between VPCs in the same or different accounts without VPC peering or Transit Gateway, providing fine-grained access control at the service level.

Best Practices

- Plan your CIDR ranges carefully: Allocate larger CIDR blocks than you think you need, and avoid overlaps with on-premises networks or future connected VPCs.
- Use three-tier subnet architecture: Separate public, private application, and private data subnets for defense in depth.
- Deploy NAT Gateways per AZ: Place a NAT Gateway in each AZ used by private subnets to avoid cross-AZ data transfer charges and eliminate single points of failure.
- Use VPC Flow Logs: Enable VPC Flow Logs to capture IP traffic information for security analysis, network troubleshooting, and compliance.
- Use Transit Gateway for complex hub-and-spoke networks: Prefer Transit Gateway over VPC peering meshes when connecting more than a handful of VPCs.
- Prefer PrivateLink for AWS service access: Use Interface VPC Endpoints (PrivateLink) to keep traffic to AWS services on the AWS network.
- Implement least-privilege security groups: Define security group rules at the individual application tier level rather than using broad CIDR-based rules.
- Separate production and non-production VPCs: Use separate VPCs, accounts, and AWS Organizations OUs for production and non-production environments.

Frequently Asked Questions

Q: How many VPCs can I create per Region?

A: By default, you can create up to five VPCs per Region per AWS account. This soft limit can be increased by submitting a service quota increase request.

Q: What is the difference between a security group and a network ACL?

A: Security groups are stateful firewalls applied at the instance level; return traffic is automatically allowed. Network ACLs are stateless, operate at the subnet level, and require explicit rules for both inbound and outbound traffic including return traffic.